La Cooperativa Agraria Agrofrutos Orgánicos Physical Export

Database model documentation

Created with Redgate Data Modeler

1. Model details

Model name: La Cooperativa Agraria Agrofrutos Orgánicos Physical Export

Version: 2.4

Database engine: MySQL

2. Tables

2.1. Table Partner

Description: Details of the affiliated producer/farmer (identity, contact information, certification status).

2.1.1. Columns

Column name	Type	Properties	Description
Member_ID	int	PK	Unique partner/product identifier
DNI_RUC	varchar(11)		Identity document or RUC
Member_Code	varchar(20)		Internal code assigned to the partner
Phone	varchar(9)	null	Contact number (see previous conversation)
Address	varchar(15)	null	Partner's address
First_Names_Last_Names	varchar(100)		Full name of the member
Certification_Status	varchar(150)		Status of your certification (valid, in process, etc.)
Document_Type_ID	int		
Document_Type_ID_Document_Type	int		Unique primary key and auto-incrementing numeric identifier for the type of identity document.

2.2. Table Parcel

Description: Plots of land belonging to a member, including their location, area, and traceability.

2.2.1. Columns

Column name	Type	Properties	Description
Plot_ID	int	PK	Unique parcel identifier
Location_Name	varchar(100)		Location name or reference

Area_Hectares	decimal(8,2)		Total area in hectares
Plot_Traceability_Code	varchar(30)		Unique traceability code
GPS_coordinates	varchar(50)	null	Location coordinates (lat, long)
Member_ID	int		Partner who owns the plot of land
Member_Member_ID	int		Unique partner/product identifier

2.3. Table Product

Description: Catalog of products handled by the company (name, variety, type of certification).

2.3.1. Columns

Column name	Type	Properties	Description
Product_ID	int	PK	Unique product identifier
Product_Name	varchar(80)		Trade name (e.g., "Kent mango")
Variety	varchar(50)	null	specific product variety
Certification_Type	varchar(50)	null	Type of certification (organic, fair trade, etc.)

2.4. Table Caliber

Description: The size/weight classification of each product (e.g., grade A, B) determines its intended use (export, local market, etc.).

2.4.1. Columns

Column name	Type	Properties	Description
Caliber_ID	int	PK	Unique caliber identifier
Caliber_Code	varchar(10)		Caliber short code (e.g., "A", "12")
Weight_Range_Grams	varchar(20)		Weight range covered (e.g., "150-200g")

Intended_Use	varchar(50)	null	Intended use (export, local market, etc.)
Product_ID	int		Product to which this caliber belongs
Product_Product_ID	int		Unique product identifier
Inventory_Stock_Inventory_Id	int		Unique identifier for the inventory record

2.5. Table Store

Description: Record each physical warehouse where the processed product is stored (location, name, capacity in m³).

2.5.1. Columns

Column name	Type	Properties	Description
Warehouse_ID	int	PK	Unique warehouse identifier
Warehouse_Name	varchar(80)		Warehouse name (e.g., "Central Warehouse")
Location	varchar(150)	null	Address or area where it is located
Capacity_m3	decimal(8,2)	null	Storage capacity in cubic meters

2.6. Table Harvest_Reception

Description: Record each harvest delivery made by a member from a specific plot (date, quality, incoming batch).

2.6.1. Columns

Column name	Type	Properties	Description
Reception_ID	int	PK	Unique receipt identifier
Delivery_Date_Time	datetime		Date and time the harvest was delivered
Entry_Batch_Code	varchar(30)		Batch code being entered
Quality_Status	varchar(30)		Quality control result

Observations	Text	null	Additional notes regarding the reception
Member_ID	int		Partner who delivered the harvest
Plot_ID	int		Plot from which it originates
User_ID	int		
Member_Member_ID	int		Unique partner/product identifier
Plot_Plot_ID	int		Unique parcel identifier
User_User_ID	int		Primary key and unique identifier for the user in the system.

2.7. Table Harvest_Receipt_Detail

Description: Break down each receipt by product and size, including the kilograms delivered and the amount to be paid.

2.7.1. Columns

Column name	Type	Properties	Description
Reception_Detail_ID	int	PK	Unique identifier for the detail
Quantity_Kilograms	decimal(8,2)		Kilograms delivered at that reception
Precio_Unidad_Establecido	decimal(8,2)	null	Agreed price per unit/kilo
Total_Payment_Amount	decimal(8,2)	null	Total amount to be paid to the partner
Reception_ID	int		Reception to which this detail belongs
Product_ID	int		Product delivered
Caliber_ID	int		Gauge of the delivered product
Harvest_Reception_Reception_ID	int		Unique receipt identifier
Caliber_Id_Caliber	int		Unique caliber identifier

2.8. Table Inventory_Stock

Description: It tracks the available stock—by lot, product, and size—in each warehouse. It serves as a "bridge" table between the physical aspect (warehouse) and the items produced (product, size).

2.8.1. Columns

Column name	Type	Properties	Description
Inventory_ID	int	PK	Unique identifier for the inventory record
Origin_Lot_Code	varchar(30)	null	Batch code from which the stock originates
Available_Quantity_in_Kilograms	decimal(8,2)		Kilograms currently available
Date_of_Last_Update	date		Last time stock was updated
Warehouse_ID	int		Warehouse where it is stored
Product_ID	int		Product to which it corresponds
Caliber_ID	int		Size of the stored product
Member_ID	int		
Reception_ID	int		
Warehouse_Warehouse_ID	int		Unique warehouse identifier
Member_Member_ID	int		Unique partner/product identifier
Harvest_Reception_Reception_ID	int		Unique receipt identifier

2.9. Table Contact

Description: Stores messages sent by visitors via the web form. Independent of the rest of the model.

2.9.1. Columns

Column name	Type	Properties	Description
Contact_ID	int	PK	Unique identifier and auto-incrementing primary key for each

			message received via the form.
Names	varchar(50)		• • Stores the name or names of the visitor who sends the message.
Surnames	varchar(50)		• • Stores the surnames of the visitor making contact.
Phone	varchar(15)		• • Records the contact phone number provided by the user.
Message	Text		
Mail	varchar(100)		• • Visitor’s email address for responding to their inquiry.
Date_Sent	datetime		It automatically records the exact date and time the message was sent.

2.10. Table Customer

Description: Record the personal, contact, and identification information of the clients or buyers who purchase the agricultural products marketed by the organization.

2.10.1. Columns

Column name	Type	Properties	Description
Customer_ID	int	PK	Unique identifier and primary key for the client.
Name	varchar(60)		Client's name, or the company name/legal business name if it is a business entity.
Last_name	varchar(80)		Client's paternal and maternal surnames.

Document_Number	varchar(15)		Official identity document number or the client’s RUC/tax ID number.
Phone	varchar(15)	null	Customer's contact phone or mobile number.
Mail	varchar(100)		Client's primary email address for communications and billing.
Address	varchar(100)	null	Client's tax address or registered address.
Creation_Date	datetime		Date and time when the client was registered in the system. (Remember to change its data type to Date, as requested by the professor.)
Document_Type_ID	int		Foreign key that references the customer's identity document type.
Document_Type_ID_Document_Type	int		Unique primary key and auto-incrementing numeric identifier for the type of identity document.

2.11. Table Document_Type

Description: Master table used to classify and standardize official identity documents (national ID, tax ID, foreigner registration card, passport) used by partners and clients within the system.

2.11.1. Columns

Column name	Type	Properties	Description
Document_Type_ID	int	PK	Unique primary key and auto-incrementing numeric identifier for the type of

			identity document.
Code	varchar(10)		Abbreviated code or standardized acronym for the document type (e.g., DNI, RUC, CE, PAS).
Name	varchar(50)		Full description or name of the identity document (e.g., National Identity Document, Taxpayer Identification Number).

2.12. Table Role

Description: Master table that defines user roles and authorization levels within the desktop application (e.g., Administrator, Collector, Partner).

2.12.1. Columns

Column name	Type	Properties	Description
Role_ID	int	PK	Unique identifier and primary key for the user role.
Name	varchar(30)		Name or title of the role assigned to the user (e.g., Administrator, Collector, Partner).
Description	varchar(150)		Optional description of the scope and privileges associated with the role in the system. (Remember to uncheck the M box for this field to make it optional.)

2.13. Table User

Description: It stores access accounts and credentials for authenticated users of the JavaFX desktop system, allowing for the management of permissions based on their assigned role.

2.13.1. Columns

Column name	Type	Properties	Description
User_ID	int	PK	Primary key and unique identifier for the user in the system.
Member_ID	int		Optional foreign key that links the user to a member in the registry, in the event that the account belongs to a farmer member. (Remember to uncheck the 'M' box for this field to make it optional.)
Role_ID	int		Foreign key that assigns the permission profile or access role to the user (e.g., Administrator, Collector, Partner).
Username	varchar(30)		The user's account name or login alias (username) on the platform.
Password_Hash	varchar(255)		Encrypted hash of the user's password for secure authentication within the system.
Registration_Date	date		Date on which the user account was created in the system.
Role_Id_Role	int		Unique identifier and

			primary key for the user role.
Member_Member_ID	int		Unique partner/product identifier

2.14. Table Plot_Product

Description: Intermediate table that links agricultural plots to the crops grown on them, recording the area sown and the date of sowing.

2.14.1. Columns

Column name	Type	Properties	Description
Plot_ID	int	PK	Foreign key and part of the composite primary key that identifies the agricultural plot.
Product_ID	int		Foreign key and part of the composite primary key that identifies the planted product. (Remember to check the PI box for this field to define the composite primary key.)
Hectares_Sown	decimal(8,2)		Land area, measured in hectares, allocated to a specific crop on the plot.
Fecha_Siembra	date		Estimated or recorded date on which the product was planted. (Remember to uncheck the M box to make this an optional field.)
Plot_Plot_ID	int		Unique parcel identifier
Product_Product_ID	int		Unique product identifier

3. References

3.1. Reference Parcela_Socio

Description: Un socio puede poseer o administrar una o varias parcelas. Cada parcela pertenece a un único socio.

Partner	0..*	Parcel
Member_ID	<->	Member_Member_ID

3.2. Reference Recepcion_Cosecha_Socio

Description: Un socio realiza uno o más entregas/recepciones de cosecha en el tiempo.

Partner	0..*	Harvest_Reception
Member_ID	<->	Member_Member_ID

3.3. Reference Recepcion_Cosecha_Parcela

Description: De una misma parcela se pueden registrar múltiples entregas de cosecha a lo largo de las campañas.

Parcel	0..*	Harvest_Reception
Plot_ID	<->	Plot_Plot_ID

3.4. Reference FK_Calibre_Product

Description: Un producto (ej. palta, mango) puede clasificarse en diferentes calibres o rangos de peso. Cada calibre pertenece de forma concreta a un solo producto.

Product	0..*	Caliber
Product_ID	<->	Product_Product_ID

3.5. Reference Detalle_Recepcion_Cosecha_Recepcion_Cosecha

Description: Una recepción de cosecha (cabecera) contiene uno o más ítems/detalles de los productos o calibres entregados en ese lote.

Harvest_Reception	0..*	Harvest_Receipt_Detail
Reception_ID	<->	Harvest_Reception_Reception_ID

3.6. Reference Detalle_Recepcion_Cosecha_Calibre

Description: Un tamaño/calibre específico puede repetirse en los detalles de distintas recepciones.

Caliber	0..*	Harvest_Receipt_Detail
Caliber_ID	<->	Caliber_Id_Caliber

3.7. Reference Calibre_Inventario_Stock

Description: El control de inventario se gestiona al nivel de detalle del calibre.

Inventory_Stock	0..*	Caliber
Inventory_ID	<->	Inventory_Stock_Inventory_Id

3.8. Reference FK_InventarioStock_Almacen

Description: Un almacén o bodega física guarda múltiples registros de existencias (stock) de diferentes productos y calibres.

Store	0..*	Inventory_Stock
Warehouse_ID	<->	Warehouse_Warehouse_ID

3.9. Reference Inventario_Stock_Socio

Partner	0..*	Inventory_Stock
Member_ID	<->	Member_Member_ID

3.10. Reference Socio_Tipo_Documento

Document_Type	0..*	Partner
Document_Type_ID	<->	Document_Type_ID_Document_Type

3.11. Reference Cliente_Tipo_Documento

Document_Type	0..*	Customer
Document_Type_ID	<->	Document_Type_ID_Document_Type

3.12. Reference Usuario_Rol

Role	0..*	User
Role_ID	<->	Role_Id_Role

3.13. Reference Parcela_Producto_Parcela

Parcel	0..*	Plot_Product
Plot_ID	<->	Plot_Plot_ID

3.14. Reference Parcela_Producto_Producto

Product	0..*	Plot_Product
Product_ID	<->	Product_Product_ID

3.15. Reference Recepcion_Cosecha_Usuario

User	0..*	Harvest_Reception
User_ID	<->	User_User_ID

3.16. Reference Inventario_Stock_Recepcion_Cosecha

Harvest_Reception	0..*	Inventory_Stock
Reception_ID	<->	Harvest_Reception_Reception_ID

3.17. Reference Usuario_Socio

Partner	0..1	User
Member_ID	<->	Member_Member_ID

4. Notes

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